

CHESAPEAKE, VA, USA

WMO: 724006

Lat: 36.666N

Lon: 76.321W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 03701

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.5	-0.4	5.4	30.8	5.2	7.2	30.6	24.6	53.7	20.7	52.2	3.6	330	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	95.0	78.2	92.2	77.4	90.4	76.7	81.5	89.5	80.3	88.2	78.9	86.6	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	152.5	85.2	78.7	149.1	84.8	76.7	139.3	83.4	45.1	89.5	43.5	87.6	42.5	86.0	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.4	16.8	14.6	DB	11.8	98.3	6.8	2.6	6.9	100.2	2.9	101.7	-0.9	103.1	-5.9	105.0
			WB	10.7	83.5	6.4	2.3	6.0	85.2	2.2	86.5	-1.4	87.8	-6.0	89.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	41.1	43.2	49.4	59.9	68.3	76.3	79.4	78.1	72.9	62.8	52.2	45.9
	DBStd	15.88	11.02	9.60	9.90	9.18	7.75	5.76	4.40	4.53	5.50	8.43	9.16	9.84
	HDD50	996	317	236	140	18	0	0	0	0	0	6	80	199
	HDD65	3258	740	613	491	205	60	2	0	0	8	146	397	596
	CDD50	4972	42	45	122	316	568	788	912	871	688	402	145	73
	CDD65	1758	1	1	8	53	163	340	447	406	246	76	12	5
	CDH74	16581	2	11	85	539	1478	3329	4611	3873	1983	577	73	20
	CDH80	6471	0	0	12	153	515	1414	2016	1601	618	135	6	1
Wind	WSAvg	5.3	6.0	6.7	6.8	6.9	5.6	4.6	4.2	3.7	4.6	4.5	4.9	5.2
Precipitation	PrecAvg	50.4	3.7	3.0	4.0	3.5	3.9	4.5	5.6	5.4	5.4	3.9	3.5	3.5
	PrecMax	63.8	6.4	7.8	9.4	6.6	5.5	11.6	9.4	12.9	15.0	8.5	10.3	7.8
	PrecMin	31.5	1.1	1.0	0.8	1.0	1.4	1.5	1.9	1.6	0.6	0.0	0.2	1.4
	PrecStd	8.7	1.5	1.7	1.9	1.5	1.2	2.3	2.1	2.8	3.5	2.0	2.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.1	75.5	81.6	90.2	93.1	98.9	98.8	97.3	91.5	88.4	81.0	76.8
		MCWB	64.6	64.6	63.0	70.9	76.5	79.2	79.7	80.0	76.2	73.9	69.2	65.4
	2%	DB	69.8	71.1	75.4	83.9	89.8	93.5	94.4	93.1	89.6	83.6	75.2	71.8
		MCWB	62.7	60.3	61.8	67.0	73.8	77.2	78.9	78.1	75.9	72.7	65.4	65.3
	5%	DB	64.0	65.9	72.1	80.9	85.7	90.8	91.4	90.8	86.2	80.8	72.1	66.3
		MCWB	58.4	58.6	60.2	65.5	71.8	75.9	77.8	77.7	74.2	70.3	64.3	62.3
	10%	DB	59.5	61.3	67.6	75.4	82.0	88.1	90.0	88.3	83.5	76.7	68.1	63.3
		MCWB	53.8	53.3	59.5	63.7	70.1	74.7	77.3	76.8	73.1	69.1	61.7	58.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	67.3	69.4	74.2	80.4	83.4	83.6	83.2	80.8	79.0	72.1	69.8
		MCDB	71.1	72.5	73.6	84.0	90.1	92.2	93.2	92.3	86.0	82.2	75.8	74.6
	2%	WB	63.7	63.9	65.9	71.1	76.3	79.9	81.4	81.3	78.6	75.4	68.6	66.2
		MCDB	66.5	67.8	72.1	79.2	85.0	89.0	90.1	89.2	83.5	80.1	73.6	69.7
	5%	WB	59.7	60.1	63.0	68.9	74.0	78.1	80.2	80.0	77.2	73.2	65.7	63.3
		MCDB	64.0	65.1	69.2	75.9	81.5	87.4	88.8	87.5	83.2	77.3	70.3	65.7
	10%	WB	54.9	54.1	59.9	66.4	72.2	76.7	78.7	78.4	75.7	70.8	63.0	59.4
		MCDB	58.2	59.7	66.4	73.5	79.8	85.3	86.9	85.5	81.5	75.2	66.6	62.7
Mean Daily Temperature Range	5% DB	MDBR	19.6	20.8	21.6	22.7	20.6	19.6	18.7	18.6	18.6	21.2	21.9	20.3
		MCDBR	25.3	26.8	27.9	27.0	24.0	22.7	20.9	21.1	22.0	23.6	24.3	21.4
		MCWBR	19.6	18.2	16.7	12.7	11.1	9.2	8.9	9.4	10.8	13.2	15.5	16.8
	5% WB	MCDBR	23.4	25.5	22.3	22.2	20.6	20.3	18.7	18.3	17.0	18.6	19.6	20.3
		MCWBR	20.6	21.0	17.0	13.3	11.1	9.1	9.0	9.2	9.2	11.8	15.1	17.7
Clear-Sky Solar Irradiance	taub	0.309	0.322	0.349	0.390	0.434	0.485	0.510	0.502	0.406	0.358	0.325	0.314	
	taud	2.532	2.490	2.444	2.342	2.248	2.160	2.104	2.111	2.376	2.500	2.538	2.549	
	Ebn at Noon	282	291	291	283	271	256	249	247	269	274	273	272	
	Edh at Noon	25	29	34	39	44	48	50	49	36	29	25	23	
All-Sky Solar Radiation	RadAvg	754	993	1321	1687	1892	2021	1922	1731	1414	1131	850	673	
	RadStd	62	106	102	124	156	125	94	102	110	131	67	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	-2.07	-2.43	N/A	N/A	N/A	N/A
Regional (52 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+194

Nomenclature: See separate page